ALGEBRA 2 · UNIT 3: QUADRATIC FUNCTIONS & COMPLEX NUMBERS

Fountain Designer

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students design water jets for a town plaza fountain. They use vertex and standard form, solve by square roots, completing the square, and the quadratic formula, use the discriminant and complex solutions to test claims about heights, solve linear-quadratic systems, and design a two-jet fountain.

Complex numbers appear naturally: asking when a jet reaches a height it never reaches gives nonreal solutions. Linear-quadratic systems model light beams and walkways.

Standards Addressed

- F-IF.8a Complete the square to show extreme values
- A-REI.4b Solve quadratics, recognizing complex solutions
- N-CN.1–2 Complex numbers and operations
- A-REI.7 Solve a linear-quadratic system

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a fountain video. Part 1 forms and key features.
Day 2	Part 2 inequalities, discriminant, complex numbers.
Day 3	Part 3 design.

Materials

- Student handout
- Graphing calculators or Desmos
- Optional: a garden hose demonstration

How the Three Versions Differ

Level	What changes
CP2	Vertex form, square roots, one inequality, and basic complex arithmetic.
CP1	Completing the square, the quadratic formula, a linear-quadratic system, and the discriminant.
Honors	Two-jet systems, compound quadratic inequalities, and complex solutions as functions of height.

Watch For (Common Misconceptions)

- Taking only the positive square root.
- Writing $i^2 = 1$.
- Thinking 'no real solution' means the problem is wrong.

Extension Ideas

- Model a real fountain from a photo.
- Use Desmos sliders to design jets.
- Research how complex numbers are used in engineering.

Answer Key – CP2

Part 1: The Main Jet

#	Task	Answer
1	Give the vertex, the axis of symmetry, and the maximum height.	Vertex (4, 8), axis x = 4, max height 8 ft
2	Find where the water lands by solving $-0.5(x - 4)^2 + 8 = 0$ with square roots.	$(x - 4)^2 = 16 \rightarrow x - 4 = \pm 4 \rightarrow x = 0$ or 8
3	Rewrite h(x) in standard form.	$-0.5(x^2 - 8x + 16) + 8 = -0.5x^2 + 4x$
4	A 5-ft-tall sign stands at x = 6. Does the water clear it? By how much?	h(6) = −0.5(4) + 8 = 6 ft, so it clears the 5-ft sign by 1 ft.

Part 2: Heights & Complex Numbers

#	Task	Answer
5	Where is the water higher than 6 ft? Write and solve a quadratic inequality.	$-0.5(x - 4)^2 + 8 > 6 \rightarrow (x - 4)^2 < 4 \rightarrow 2 < x < 6$
6	A designer claims the jet reaches 9 ft. Solve h(x) = 9. What do the solutions tell you?	$(x - 4)^2 = -2 \rightarrow x = 4 \pm i\sqrt{2}$. The solutions are not real, so the jet never reaches 9 ft.
7	Simplify $(3 + 2i) + (1 - 5i)$ and $(2 + i)(3 - i)$.	$4 - 3i$; $6 - 2i + 3i - i^2 = 7 + i$
8	Explain why $i^2 = -1$ means $\sqrt{(-2)} = i\sqrt{2}$.	$\sqrt{(-2)} = \sqrt{(-1)} \cdot \sqrt{2} = i\sqrt{2}$

Part 3: Design a Jet

#	Task	Answer
9	Write your jet in vertex form $h(x) = a(x - h)^2 + k$.	Answers vary.
10	Find where it lands and its maximum height.	Answers vary.
11	Show that it can never reach 9 ft by solving h(x) = 9.	Answers vary.

Answer Key – CP1

Part 1: Two Jets

#	Task	Answer
1	Complete the square to write s(x) in vertex form. Give its maximum height.	$s(x) = -0.5(x^2 - 6x) + 1.5 = -0.5(x - 3)^2 + 6$; max 6 ft at x = 3
2	Find where the side jet lands (exact and decimal).	$(x - 3)^2 = 12 \rightarrow x = 3 + 2\sqrt{3} \approx 6.46$ ft (reject $3 - 2\sqrt{3}$)
3	Which jet goes higher? Which lands farther away?	The main jet (8 ft) goes higher and lands farther (8 ft).

Part 2: Claims & Systems

#	Task	Answer
4	Use the discriminant to decide if the side jet ever reaches 7 ft. If not, give the complex solutions.	$-0.5x^2 + 3x - 5.5 = 0$: $b^2 - 4ac = 9 - 11 = -2 < 0$, so never. $x = 3 \pm i\sqrt{2}$.
5	A light beam follows y = 0.5x + 1. Where does it meet the main jet? Give exact and decimal answers.	$-0.5x^2 + 4x = 0.5x + 1 \rightarrow x^2 - 7x + 2 = 0 \rightarrow x = (7 \pm \sqrt{41}) \div 2 \approx 0.30$ and 6.70
6	Where is the main jet above 6 ft? Solve the inequality.	$2 < x < 6$
7	Simplify $(4 + i\sqrt{2})(4 - i\sqrt{2})$ and explain why the answer is real.	$16 - (i\sqrt{2})^2 = 16 + 2 = 18$. Conjugates multiply to a real number.

Part 3: Design a Jet

#	Task	Answer
8	Write the equation and complete the square to verify the peak.	Answers vary.
9	Use the quadratic formula to find the landing point.	Answers vary.
10	Pick a height your jet never reaches and show the solutions are complex.	Answers vary.

Answer Key – Honors

Part 1: Two Crossing Jets

#	Task	Answer
1	Find where the two jets cross and the height there.	$-0.5(x-4)^2 = -0.5(x-6)^2 \rightarrow x = 5$; height 7.5 ft
2	Find where both jets are above 7 ft at the same time.	$h > 7$: $4 - \sqrt{2} < x < 4 + \sqrt{2}$. $g > 7$: $6 - \sqrt{2} < x < 6 + \sqrt{2}$. Both: $6 - \sqrt{2} < x < 4 + \sqrt{2}$ (about 4.59 to 5.41).
3	Write g(x) as a transformation of h(x).	$g(x) = h(x-2)$: shift right 2.

Part 2: Complex Heights

#	Task	Answer
4	Solve $h(x) = k$ for x in terms of k.	$x = 4 \pm \sqrt{(2(8-k))}$
5	For which k are the solutions real and different, equal, or complex? Interpret each case.	$k < 8$: two real; $k = 8$: one (the vertex); $k > 8$: complex (the water never reaches k).
6	For $k = 10$, find the solutions and verify that their sum and product match the coefficients of the equation.	$x = 4 \pm i\sqrt{4} = 4 \pm 2i$. Equation $0.5x^2 - 4x + 10 = 0$, i.e., $x^2 - 8x + 20 = 0$: sum 8 ✓, product 16 + 4 = 20 ✓.
7	Simplify $(3 - 2i) \div (1 + i)$ by multiplying by the conjugate.	$(3 - 2i)(1 - i) \div 2 = (3 - 5i + 2i^2) \div 2 = (1 - 5i) \div 2 = 0.5 - 2.5i$

Part 3: Design a Two-Jet Fountain

#	Task	Answer
8	Write both jets and find where they cross.	Answers vary.
9	Prove the max-height rule with a discriminant.	Answers vary.
10	Present the design with a graph.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Fountain design meets every requirement, with complex or real solutions interpreted correctly. Goes beyond the minimum with a creative, realistic choice.	Fountain design meets every requirement, with complex or real solutions interpreted correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Proposal includes each jet's equation in two forms, the key features, and solutions to the design questions. Polished, organized, and visually engaging.	Proposal includes each jet's equation in two forms, the key features, and solutions to the design questions. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.